

Vol 1 Chapter 9 — Scattering and the S-Matrix

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Chapter 9 — Scattering and the S-Matrix

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Most of what experimental particle physics measures is scattering — collisions in which incoming particles meet, interact, and emerge as outgoing particles. The theoretical apparatus is the S-matrix, the unitary operator from free in-states to free out-states. Standard QFT develops it via the LSZ reduction formula, perturbative Feynman diagrams, and renormalized amplitude calculations.

BST's substrate framework offers a parallel apparatus with structural advantages. The substrate Hilbert space $H^2(D_{IV}^5)$ replaces Fock space; the Bergman reproducing kernel replaces the Feynman propagator (Volume 1 Chapter 2 §2.4); the substrate-cycle commitment phases of Volume 0 Chapter 3 supply natural in/out state preparation. The S-matrix is unitary, ultraviolet-complete by substrate structure, and computable on the substrate's per-tick $GF(128)^k$ discretization without standard-QFT renormalization machinery.

9.1 In-states and out-states from substrate cycle

Standard QFT defines in/out states by adiabatic switching: H_0 in the asymptotic past, H during interaction, H_0 in the asymptotic future. Subtleties occupy several chapters of standard texts.

In BST, in/out states are defined by the substrate commitment cycle. In-state: substrate state at Zone 1 (absorption — received from environment). Out-state: substrate state at Zone 4 (active edge emission — broadcast to environment). The S-matrix integrates across the cycle:

$$\hat{S} = \hat{U}_{Z_4 \leftarrow Z_3} \cdot \hat{P}_{Z_3} \cdot \hat{U}_{Z_3 \leftarrow Z_2} \cdot \hat{U}_{Z_2 \leftarrow Z_1},$$

with each $\hat{U}$ a per-zone substrate evolution and $\hat{P}_{Z_3}$ the Bergman-kernel projector (Volume 1 Chapter 7 §7.4). Unitary on $H^2(D_{IV}^5)$ by the cycle's structural unitarity.

The adiabatic-switching subtlety disappears. The substrate has a definite cycle; in/out separation is the cycle's natural structure.

9.2 LSZ reduction on the substrate

Standard LSZ expresses S-matrix elements as products of propagators times amputated Green's functions. Lyra T2457's substrate identification gives the Bergman-kernel realization:

$$\langle z_f | \hat{S} | z_i \rangle = K_B(z_f, \bar{z}_i) \cdot (\text{commitment-phase corrections}),$$

with leading term the Bergman kernel and higher orders from substrate-cycle structure. For tree-level scattering, direct kernel evaluation. For higher orders, finite sums over K-type Casimir eigenspaces (Volume 1 Chapter 10 §10.1 ultraviolet completeness).

The structural machinery is in place; explicit multi-loop QED amplitudes on the substrate side are active research.

9.3 The optical theorem

Unitarity of $\hat{S}$ implies the optical theorem (total cross-section relates to imaginary part of forward elastic amplitude). On the substrate, unitarity is structural — the commitment cycle is unitary (Born=Bergman), so $\hat{S}^\dagger \hat{S} = \hat{1}$ on $H^2(D_{IV}^5)$ automatically. The optical theorem holds in substrate-derived form, with the forward elastic amplitude's imaginary part computed as Bergman-kernel evaluation on coincident arguments.

The substrate-side version preserves the standard relation while inheriting substrate ultraviolet completeness. No divergent integrals to regulate.

9.4 What comes next

Chapter 10 explains why BST does not need standard renormalization. Chapter 11 closes the volume with the consolidated observables reference. Volume 2 begins particle-physics applications.

Where to look this up: Substrate scattering anchor T2438 + T2457. For standard-QFT scattering: Peskin & Schroeder Chapters 4 and 7. Optical theorem: Weinberg, *Quantum Theory of Fields*, Volume 1, Chapter 3. Multi-loop substrate-side calculations: Elie K52a Sessions 10+ rail.